Slide 1: Course Opening: Development Environment

Goal: Set clear outcomes and class flow.

- What students will build
- Tools used: VS Code, Python, Git, GitHub
- Expected output at the end of class

Practice: Quick check: each student shares their OS and Python version.

Slide 2: Create Virtual Environment

Goal: Create isolated project dependencies.

- Run: python -m venv .venv
- Activate environment (PowerShell/CMD/macOS/Linux)
- Confirm with: python -c "import sys; print(sys.executable)"

Practice: Students create and activate .venv in pairs.

Slide 3: Select Interpreter and Kernel

Goal: Bind VS Code and notebooks to the correct environment.

- Python: Select Interpreter
- Notebook kernel selection
- Verify interpreter path shows .venv

Practice: Students switch interpreter and run a test import.

Slide 4: Install Dependencies Professionally

Goal: Install only what is needed and keep reproducibility.

- Install: pip install pandas matplotlib ipykernel
- Freeze: pip freeze > requirements.txt
- Restore: pip install -r requirements.txt

Practice: Students install one package and validate import.

Slide 5: Verify Environment Health

Goal: Detect interpreter/package mismatch fast.

- Check sys.executable
- Import required libraries
- Run one mini script or notebook cell

Practice: Troubleshoot one intentional error (missing package).

Slide 6: Git Daily Workflow

Goal: Build reliable commit habits.

- git status -> git add -> git commit -> git push
- Use small commits with clear messages
- Keep .venv and cache files in .gitignore

Practice: Students commit one small change and push.

Slide 7: RECOMMENDED PRACTICE SLIDE: End-to-End Drill

Goal: Consolidate all steps in one hands-on activity.

- Create .venv
- Select interpreter/kernel
- Install one package
- Run verification cell
- Commit and push result

Practice: Time-boxed 15-minute challenge with checklist review.